PLRG
Programming Language
Research Group

JISET: JavaScript IR-based Semantics Extraction Toolchain

Jihyeok Park, Jihee Park, Seungmin An, Sukyoung Ryu

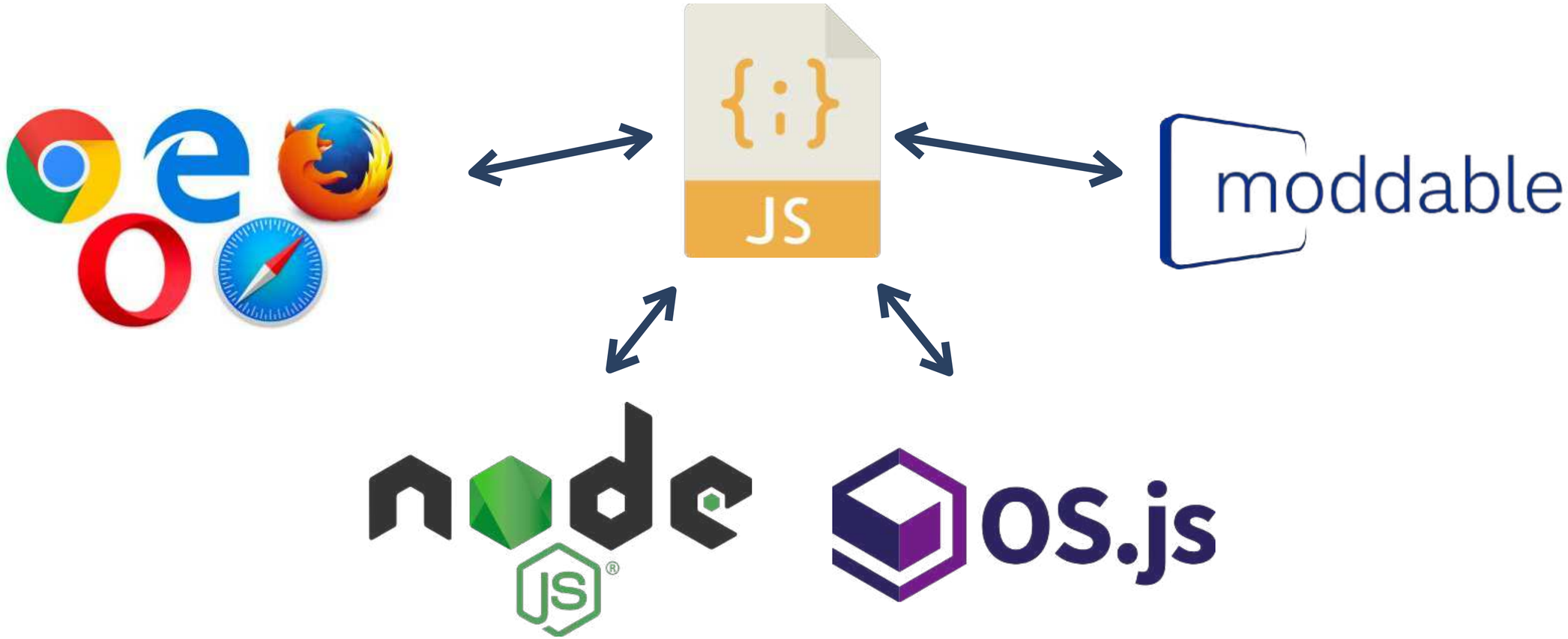
PLRG @ KAIST

In Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering (ASE'20)

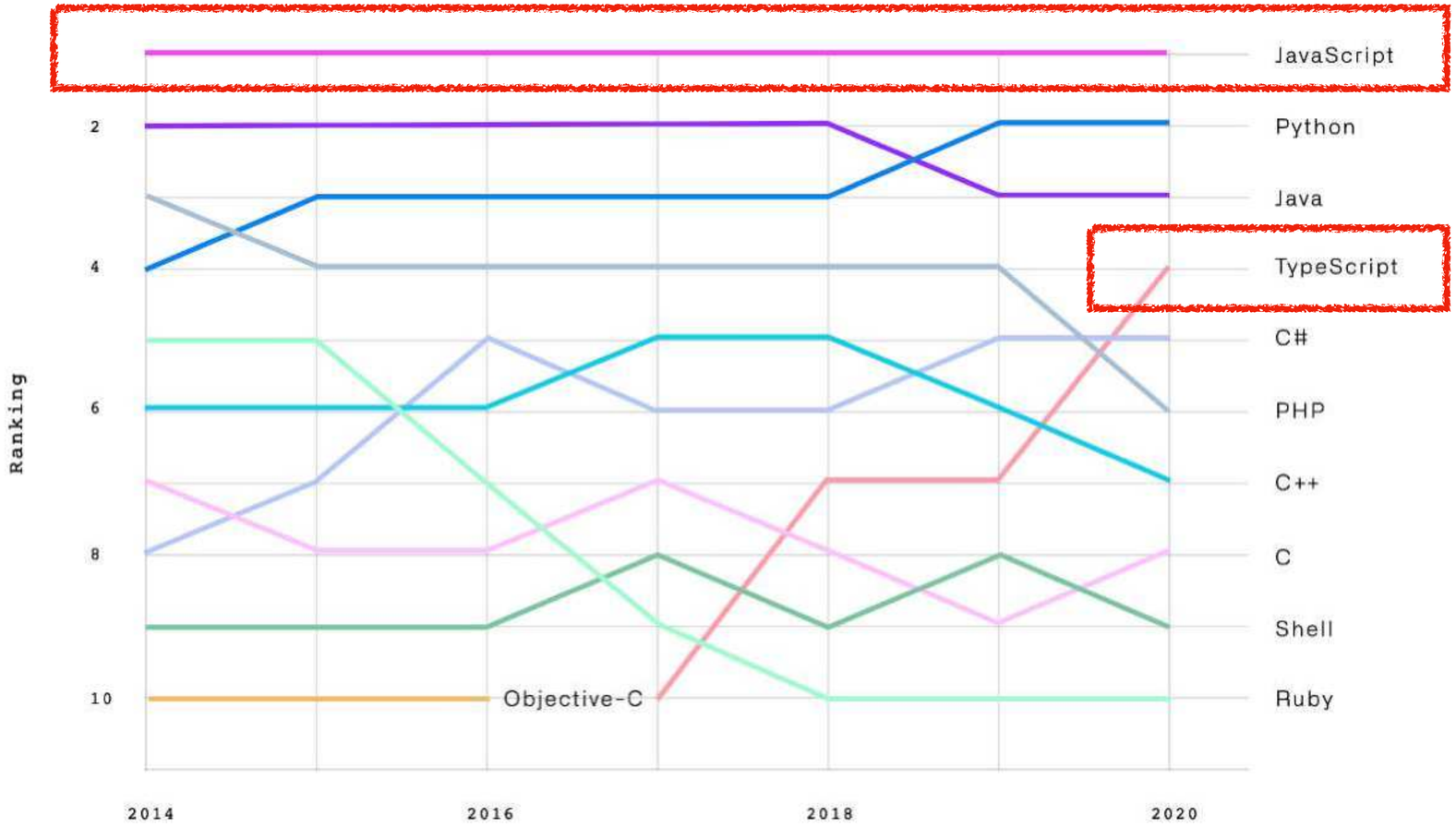
2021 한국컴퓨터종합학술대회 (KCC2021) Top Conference 세션

June 23, 2021

JavaScript is Everywhere



JavaScript is Dominating



https://octoverse.github.com/

JavaScript Complex Semantics

```
function f(x) { return x == !x; }
```

Always return `false`?

JavaScript Complex Semantics

```
function f(x) { return x == !x; }
```

Always return `false`?

NO!!

```
f([]) -> [] == ![]
      -> [] == false
      -> +[] == +false
      -> 0 == 0
      -> true
```

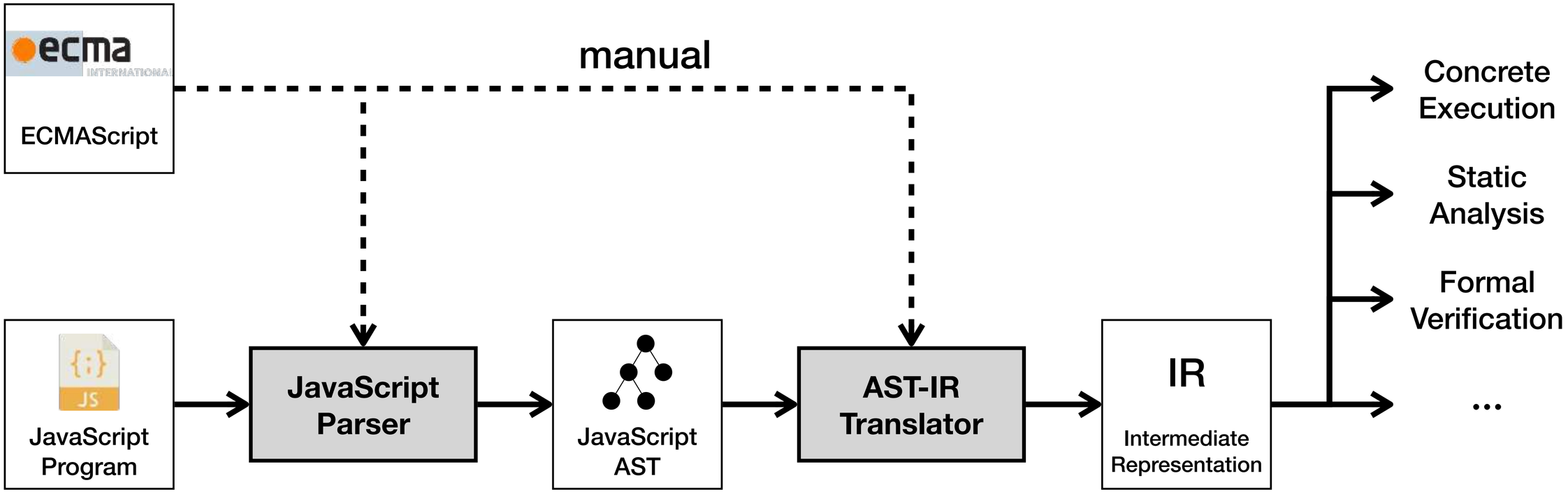
ECMAScript: JavaScript Specification



> The standards for JavaScript are the ECMAScript Language Specification (ECMA-262) and the ECMAScript Internationalization API specification (ECMA-402). The JavaScript documentation throughout MDN is based on the latest draft versions of ECMA-262 and ECMA-402. And in cases where some proposals for new ECMAScript features have already been implemented in browsers, documentation and examples in MDN articles may use some of those new features.

https://developer.mozilla.org/en-US/docs/Web/JavaScript

IR-based Semantics Extraction



IR-based Semantics Extraction

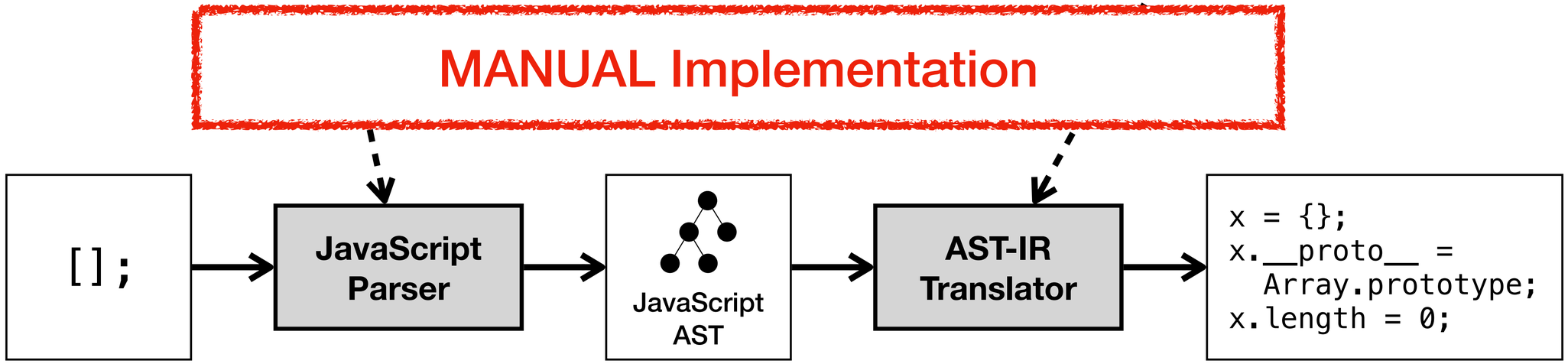
ArrayLiteral[Yield, Await] :
[*Elision*opt]
[*ElementList*[?Yield, ?Await]]
[*ElementList*[?Yield, ?Await] , *Elision*opt]

The production of *ArrayLiteral* in ES10

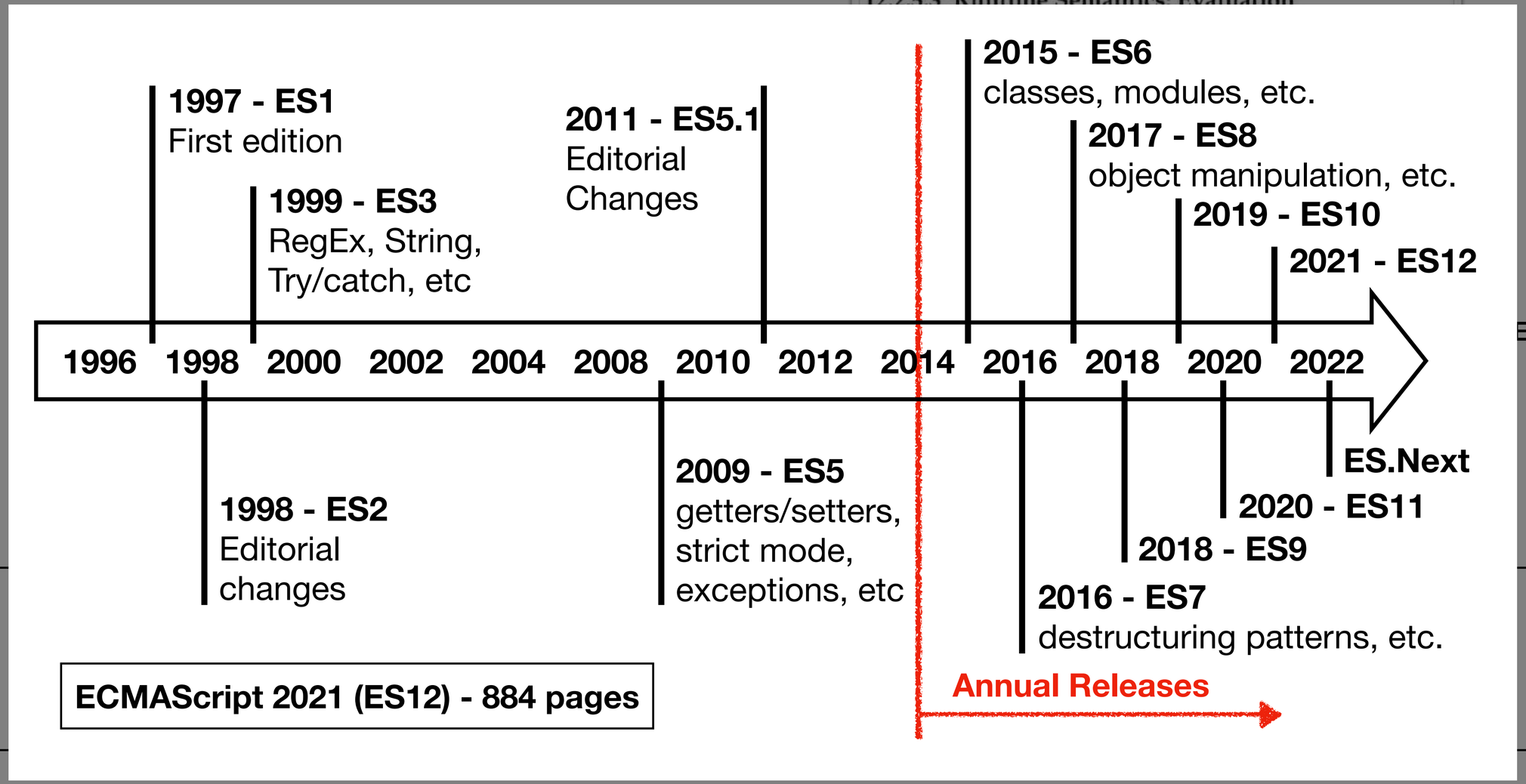
12.2.5.3 Runtime Semantics: Evaluation
ArrayLiteral : [*Elision*]

1. Let *array* be ! ArrayCreate(0).
2. Let *pad* be the ElisionWidth of *Elision*; if *Elision* is not present, use the numeric value zero.
3. Perform Set(*array*, "length", ToUint32(*pad*), **false**).
4. NOTE: The above Set cannot fail because of the nature of the object returned by ArrayCreate.
5. Return *array*.

The Evaluation algorithm for the first alternative of *ArrayLiteral* in ES10



IR-based Semantics Extraction



Our Approach

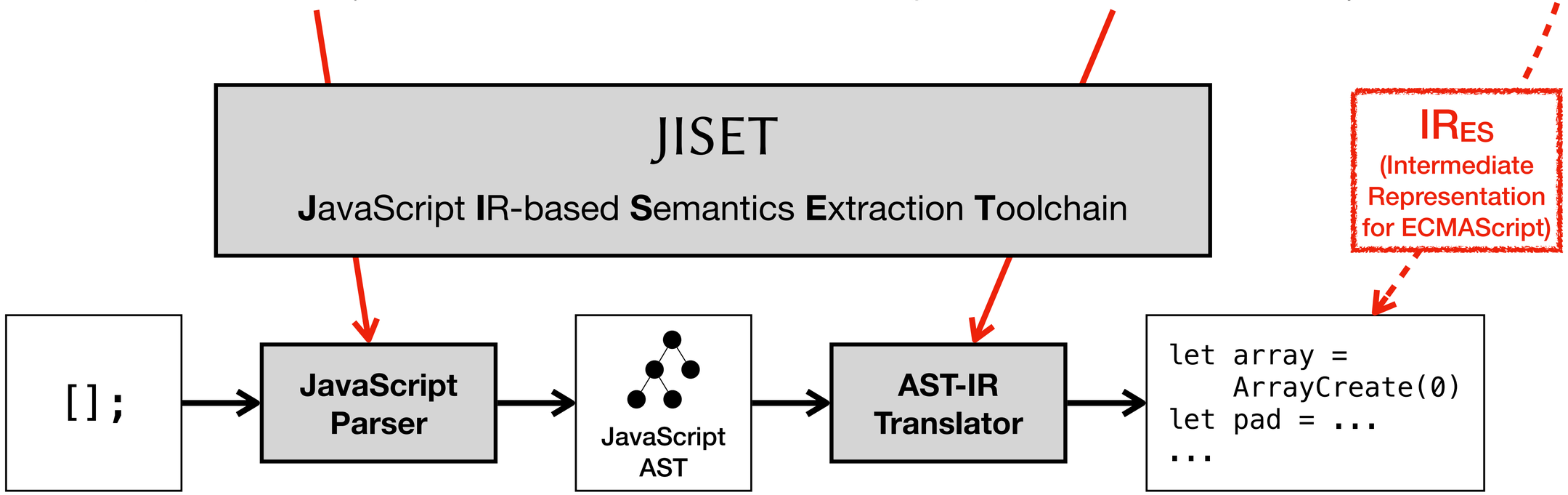
ArrayLiteral[Yield, Await] :
[*Elision*opt]
[*ElementList*[?Yield, ?Await]]
[*ElementList*[?Yield, ?Await] , *Elision*opt]

The production of *ArrayLiteral* in ES10

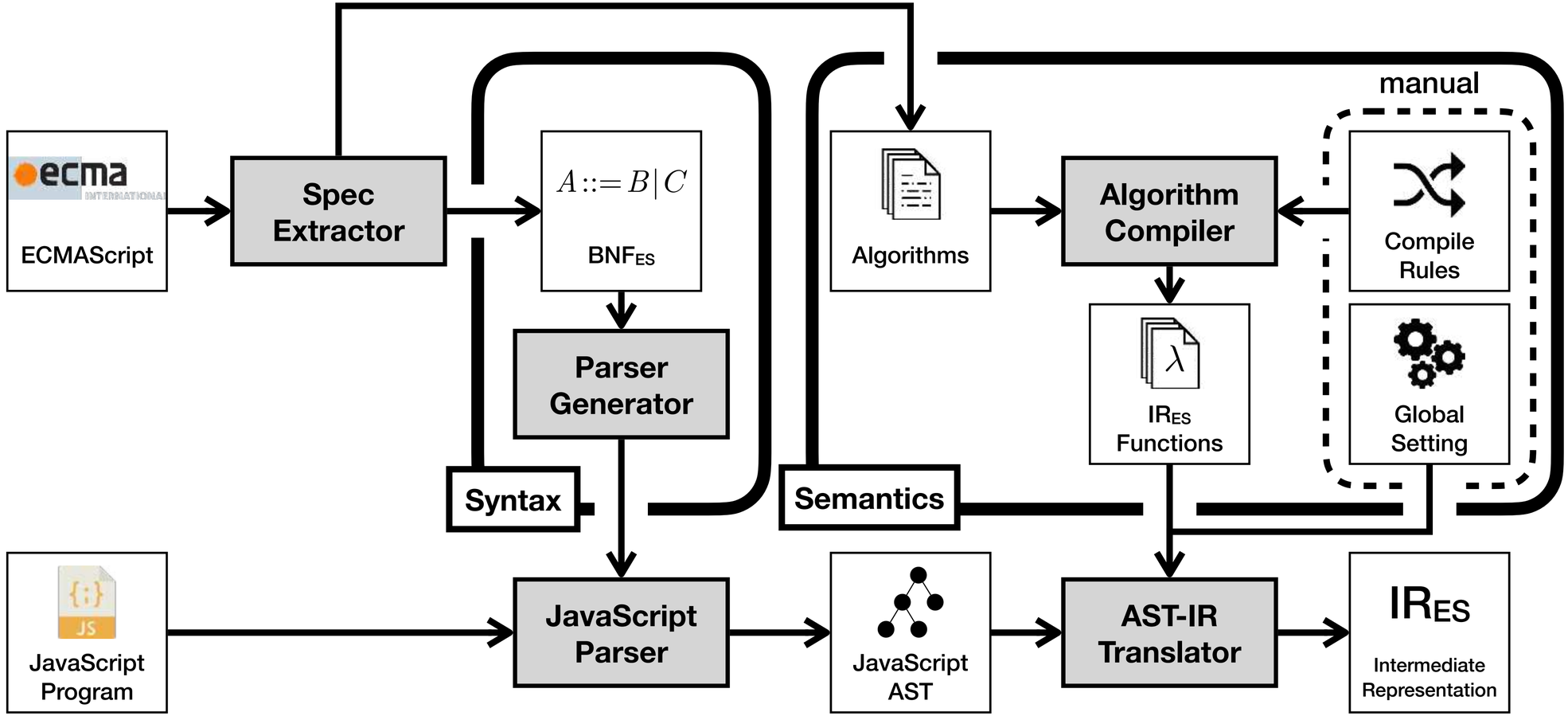
12.2.5.3 Runtime Semantics: Evaluation
ArrayLiteral : [*Elision*]

1. Let *array* be ! ArrayCreate(0).
2. Let *pad* be the ElisionWidth of *Elision*; if *Elision* is not present, use the numeric value zero.
3. Perform Set(*array*, "length", ToUint32(*pad*), **false**).
4. NOTE: The above Set cannot fail because of the nature of the object returned by ArrayCreate.
5. Return *array*.

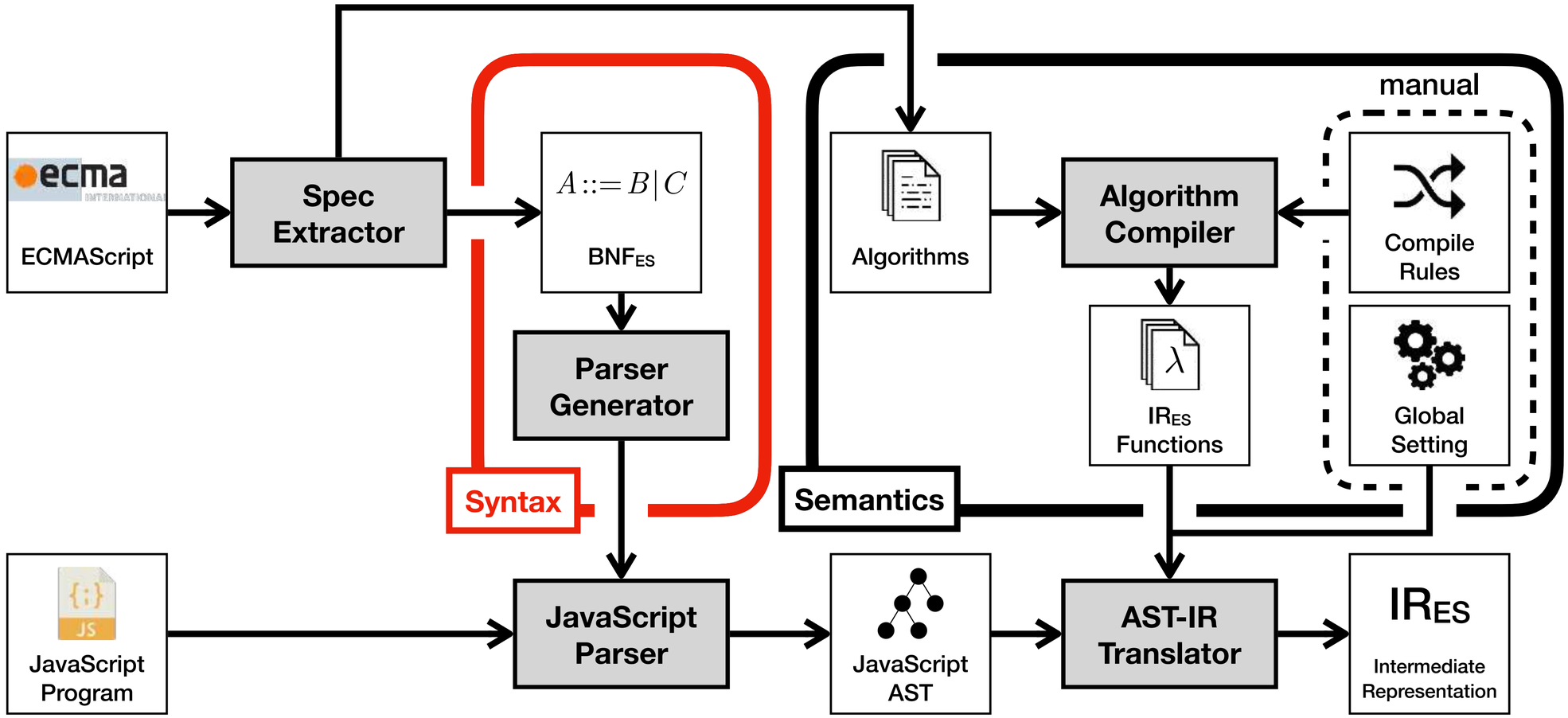
The Evaluation algorithm for the first alternative of *ArrayLiteral* in ES10



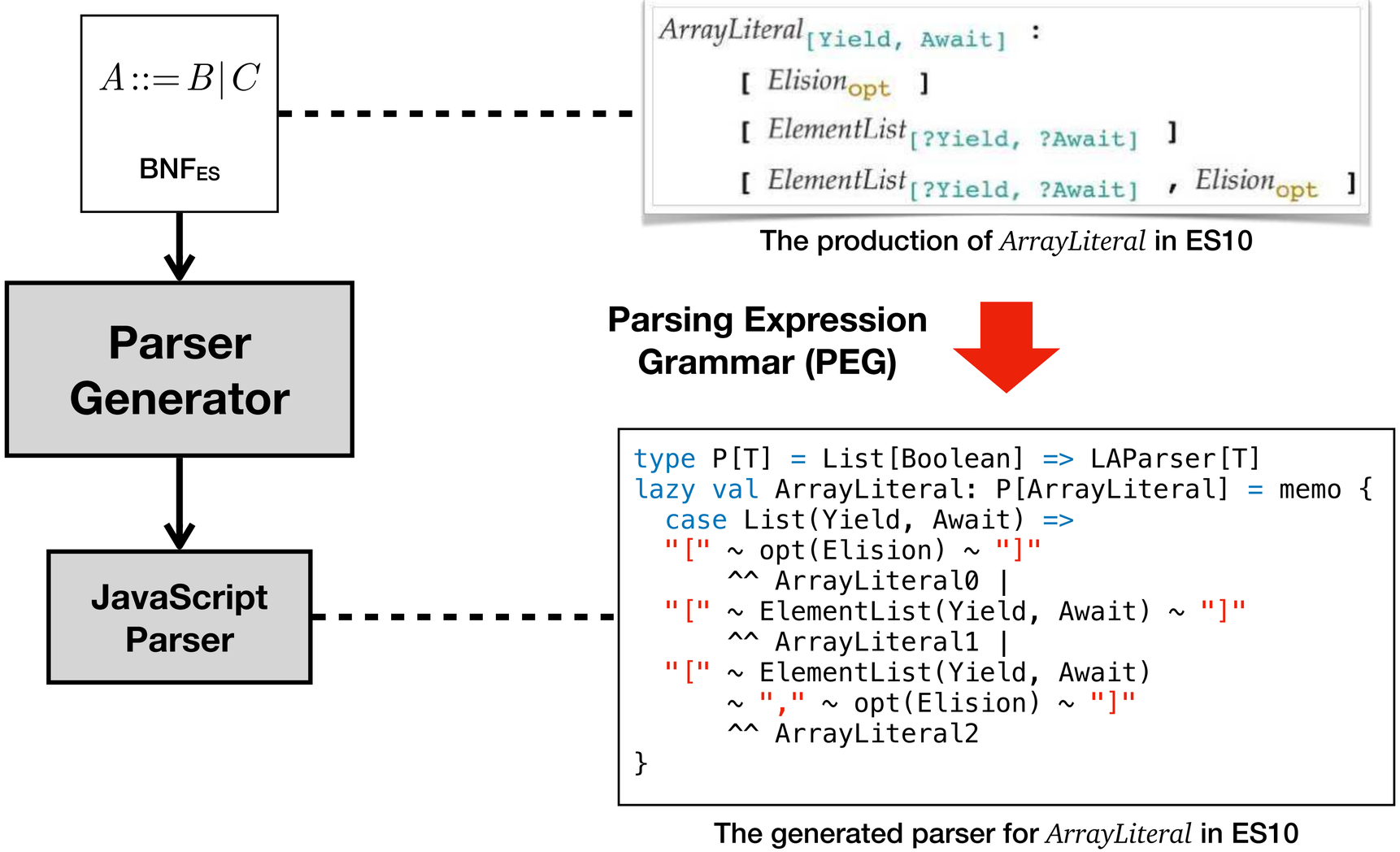
Overall Structure of JISET



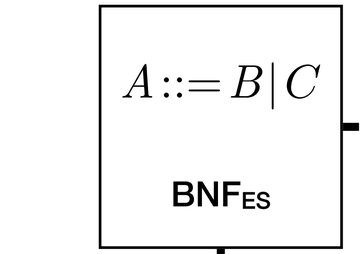
Overall Structure of JISET

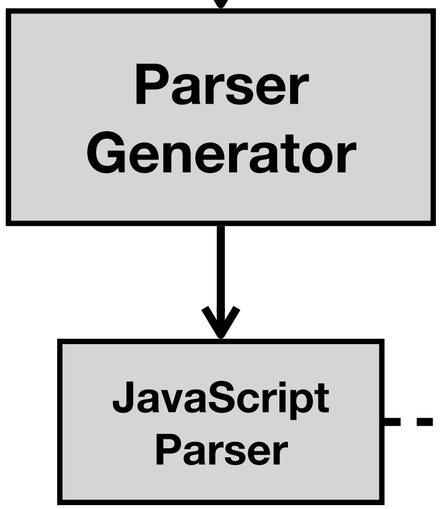


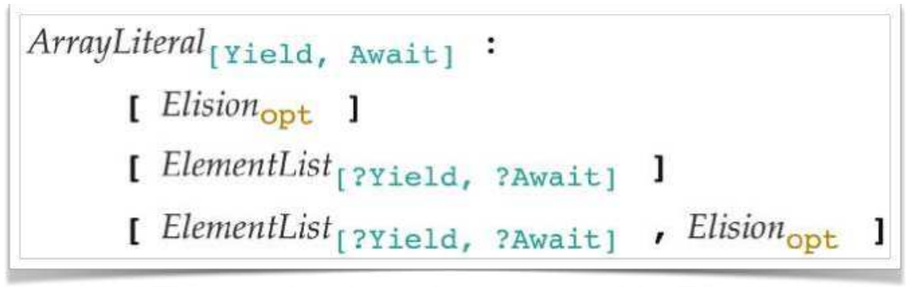
Syntax - Parser Synthesis



Syntax - Parser Synthesis







The production of *ArrayLiteral* in ES10

Parsing Expression Grammar (PEG) → **+ Lookahead Parsing**

```
type P[T] = List[Boolean] => LAParser[T]
lazy val ArrayLiteral: P[ArrayLiteral] = memo {
  case List(Yield, Await) =>
  "[" ~ opt(Elision) ~ "]"
      ^^ ArrayLiteral0 |
  "[" ~ ElementList(Yield, Await) ~ "]"
      ^^ ArrayLiteral1 |
  "[" ~ ElementList(Yield, Await)
      ~ "," ~ opt(Elision) ~ "]"
      ^^ ArrayLiteral2
}
```

The generated parser for *ArrayLiteral* in ES10

Syntax - Lookahead Parsing

$$
\begin{aligned}
\mathbf{first}_\alpha(s_1 \cdots s_n) &= \mathbf{first}_s(s_1) :+ \mathbf{first}_s(s_2 \cdots s_n) \\
&\quad \text{where } x :+ y = \begin{cases} x \cup y & \text{if } \circ \in x \\ x & \text{otherwise} \end{cases} \\
\mathbf{first}_s(\epsilon) &= \{\circ\} \\
\mathbf{first}_s(\mathsf{a}) &= \{\mathsf{a}\} \\
\mathbf{first}_s(A(a_1, \cdots, a_k)) &= \mathbf{first}_\alpha(\alpha_1) \cup \cdots \cup \mathbf{first}_\alpha(\alpha_n) \\
&\quad \text{where } A(a_1, \cdots, a_k) = \alpha_1 \mid \cdots \mid \alpha_n \\
\mathbf{first}_s(s?) &= \mathbf{first}_s(s) \cup \{\circ\} \\
\mathbf{first}_s(+s) &= \mathbf{first}_s(s) \\
\mathbf{first}_s(-s) &= \{\circ\} \\
\mathbf{first}_s(s \setminus s') &= \mathbf{first}_s(s) \\
\mathbf{first}_s(\langle \neg \mathsf{LT} \rangle) &= \{\circ\}
\end{aligned}
$$

Algorithm for first tokens of BNF$_{ES}$

Algorithm for lookahead parsing

$$
\begin{aligned}
(s_1 \cdots s_n)[L] &= s_1[\mathbf{first}_s(s_2 \cdots s_n) :+ L]\ (s_1 \cdots s_n)[L] \\
\epsilon[L] &= +\mathbf{get}_s(L) \\
\mathsf{a}[L] &= \mathsf{a}\ + \mathbf{get}_s(L) \\
A(a_1, \cdots, a_k)[L] &= \alpha_1[L] \mid \cdots \mid \alpha_n[L] \\
&\quad \text{where } A(a_1, \cdots, a_k) = \alpha_1 \mid \cdots \mid \alpha_n \\
s?[L] &= s[L] \mid \epsilon[L] \\
(\pm s)[L] &= \pm(s[L]) \\
(s \setminus s')[L] &= s[L] \setminus s' \\
\langle \neg \mathsf{LT} \rangle &= \langle \neg \mathsf{LT} \rangle\ + \mathbf{get}_s(L)
\end{aligned}
$$

Syntax - Evaluation

Version	ES7	ES8	ES9	ES10	Average
# Lexical productions	78	78	78	81	78.75
# Syntactic productions	157	167	167	174	166.25

Old version	ES7	ES8	ES9	Average
New version	ES8	ES9	ES10	
Δ # Lexical productions	3	5	6	4.67
Δ # Syntactic productions	140	15	8	54.33

Syntax - Evaluation

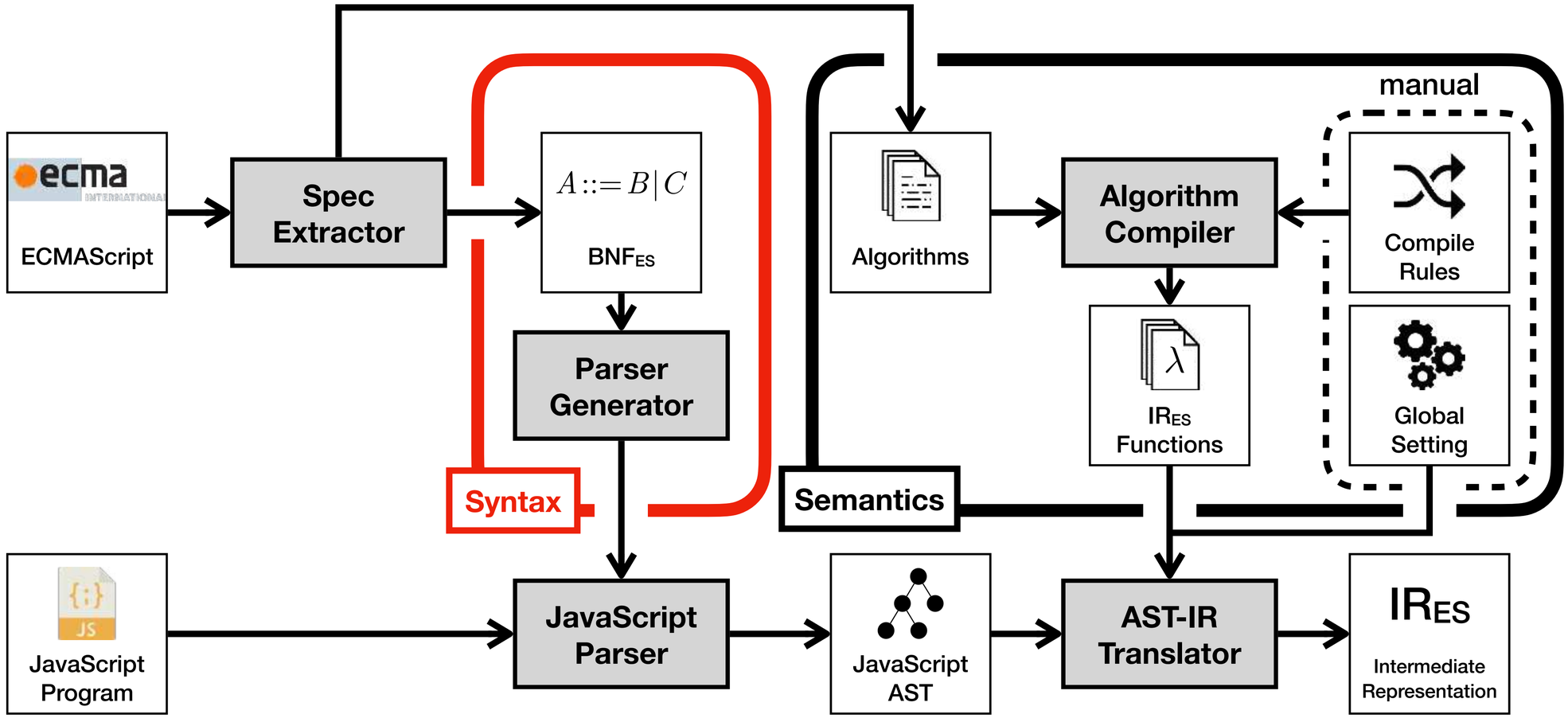
All Success!!

Version	ES7	ES8	ES9	ES10	Average
# Lexical productions	78	78	78	81	78.75
# Syntactic productions	157	167	167	174	166.25

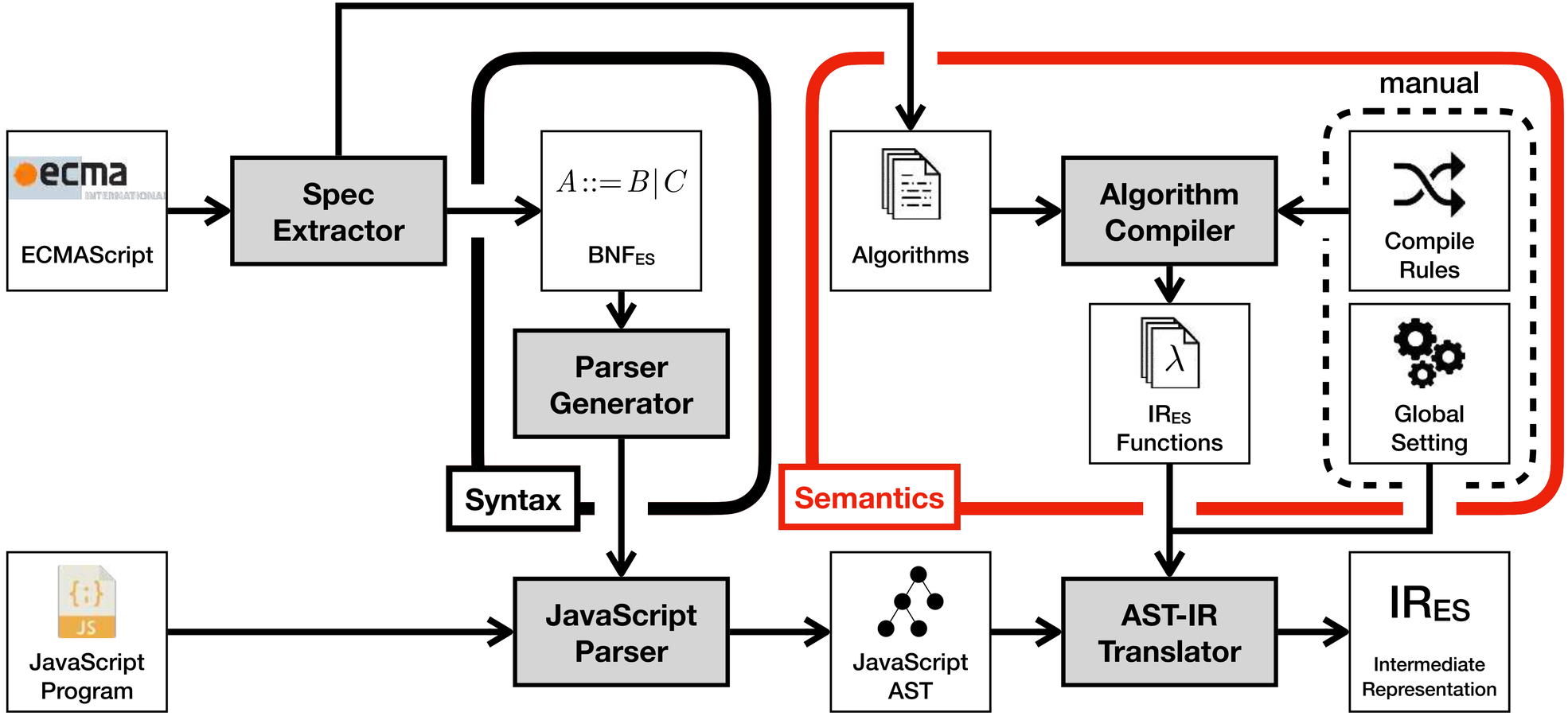
Pass all parsing tests
in Test262

Old version	ES7	ES8	ES9	Average
New version	ES8	ES9	ES10	
Δ # Lexical productions	3	5	6	4.67
Δ # Syntactic productions	140	15	8	54.33

Overall Structure of JISET



Overall Structure of JISET



Semantics - Algorithm Compilation

12.2.5.3 Runtime Semantics: Evaluation

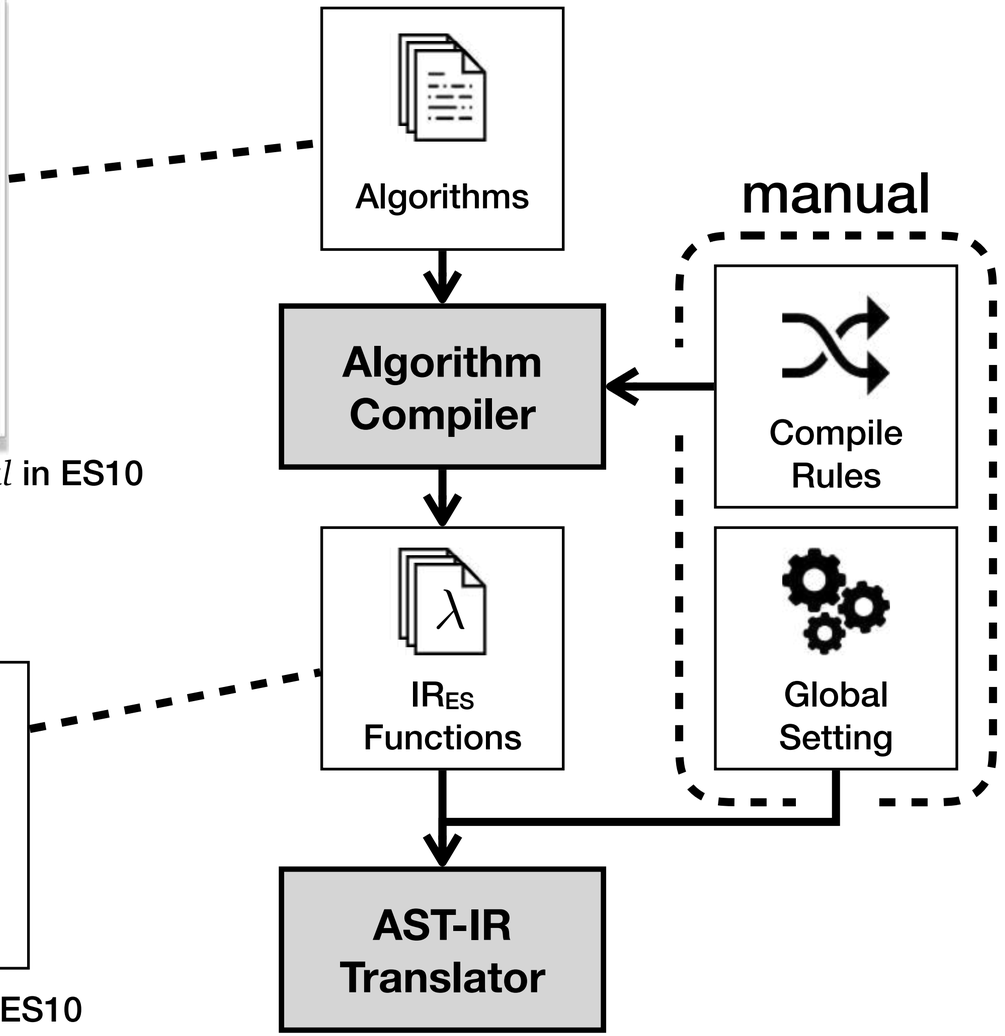
ArrayLiteral : [*Elision*]

1. Let *array* be ! ArrayCreate(0).
2. Let *pad* be the ElisionWidth of *Elision*; if *Elision* is not present, use the numeric value zero.
3. Perform Set(*array*, "length", ToUint32(*pad*), **false**).
4. NOTE: The above Set cannot fail because of the nature of the object returned by ArrayCreate.
5. Return *array*.

The Evaluation **algorithm for the first alternative of** *ArrayLiteral* **in ES10**

```
"ArrayLiteral0.Evaluation" (Elision) => {
  let array = ! (ArrayCreate 0)
  if (= Elision absent) let pad = 0
  else let pad = Elision.ElisionWidth
  (Set array "length" (ToUint32 pad) false)
  return array
}
```

The IR_{ES} function of the first alternative of *ArrayLiteral* **in ES10**



Semantics - Algorithm Compilation

12.2.5.3 Runtime Semantics: Evaluation

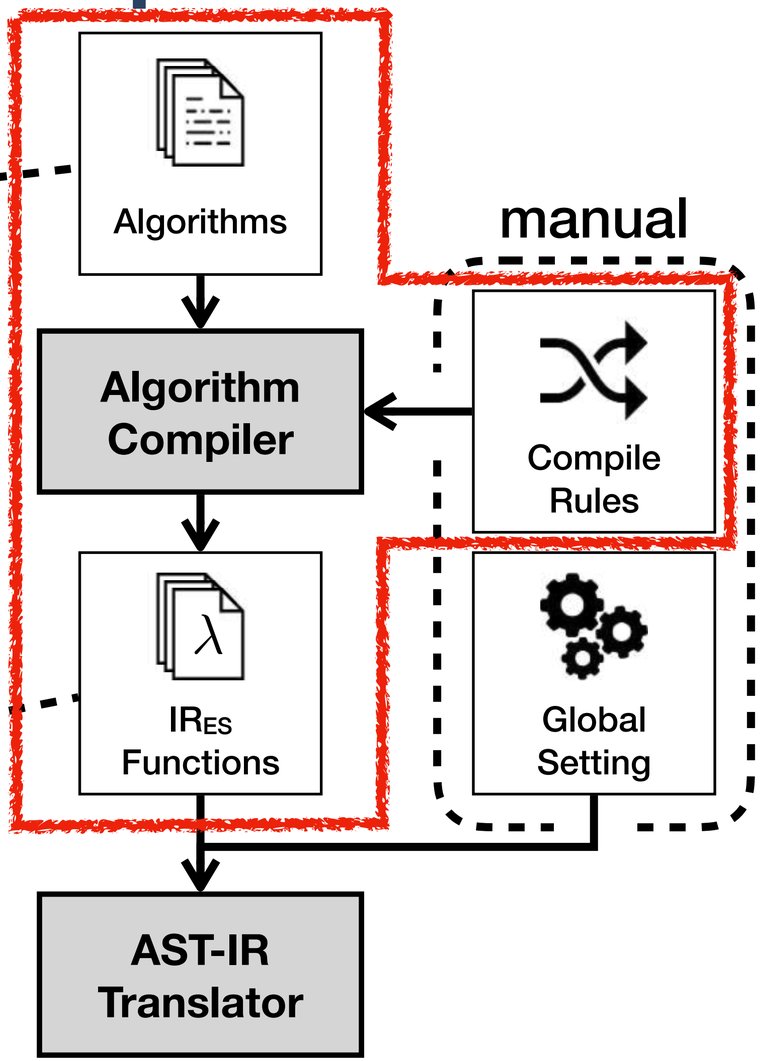
ArrayLiteral : [*Elision*]

1. Let *array* be ! ArrayCreate(0).
2. Let *pad* be the ElisionWidth of *Elision*; if *Elision* is not present, use the numeric value zero.
3. Perform Set(*array*, "length", ToUint32(*pad*), **false**).
4. NOTE: The above Set cannot fail because of the nature of the object returned by ArrayCreate.
5. Return *array*.

The Evaluation **algorithm for the first alternative of** *ArrayLiteral* **in ES10**

```
"ArrayLiteral0.Evaluation" (Elision) => {
  let array = ! (ArrayCreate 0)
  if (= Elision absent) let pad = 0
  else let pad = Elision.ElisionWidth
  (Set array "length" (ToUint32 pad) false)
  return array
}
```

The IR_{ES} function of the first alternative of *ArrayLiteral* **in ES10**



Parsing rules Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ . ^^ ILet

E = // expressions
  ! E                  ^^ EAbruptCheck |
  str ~ ( ~ E ~ )      ^^ ECall        |
  num                  ^^ _.toDouble
```

Simplified compile rules

Let *array* be ! ArrayCreate (0) .

Parsing rules Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ . ^^ ILet

E = // expressions
  ! E                  ^^ EAbruptCheck |
  str ~ ( ~ E ~ )      ^^ ECall        |
  num                  ^^ _.toDouble
```

Simplified compile rules



Parsing rules

```
S = // statements
  Let ~ V ~ be ~ E ~ .

E = // expressions
  ! E
  str ~ ( ~ E ~ )
  num
```

Conversion Rules

```
^^ ILet


^^ EAbruptCheck |
^^ ECall        |
^^ _.toDouble
```

Simplified compile rules

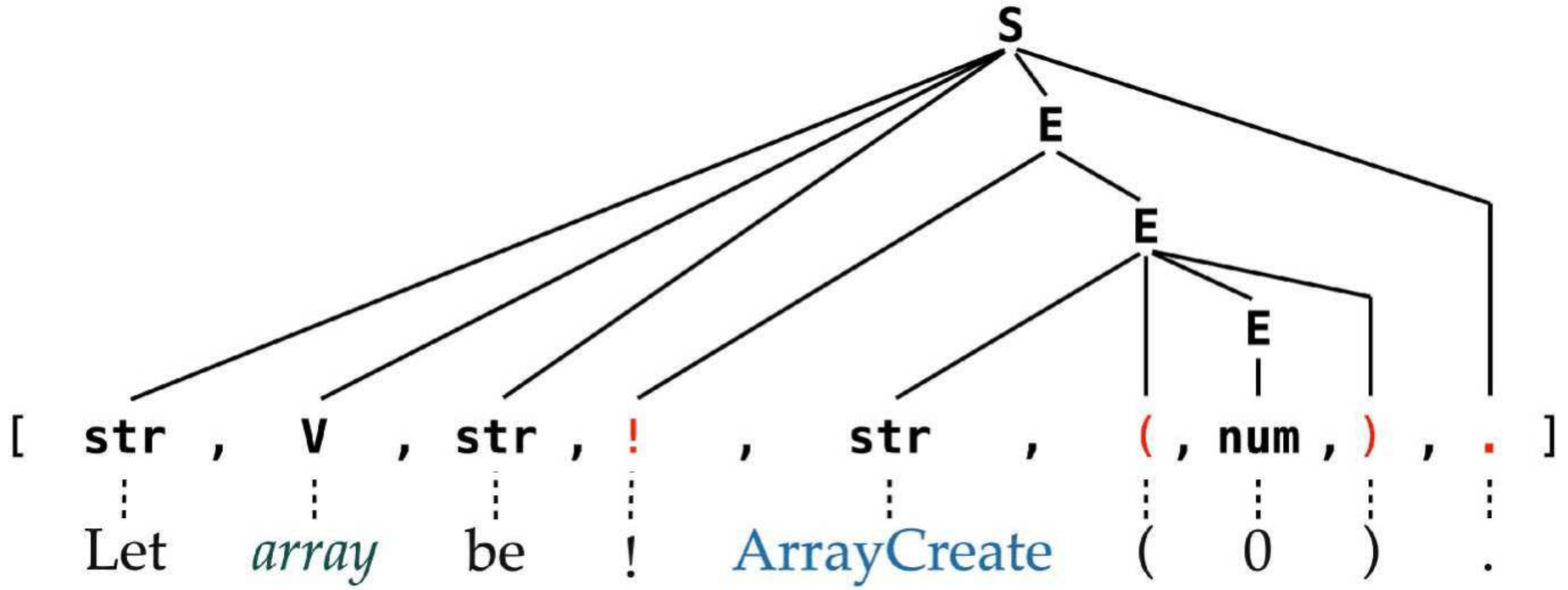


Parsing rules | Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ .   ^^ ILet

E = // expressions
  ! E                    ^^ EAbruptCheck |
  str ~ ( ~ E ~ )        ^^ ECall        |
  num                    ^^ _.toDouble
```

Simplified compile rules

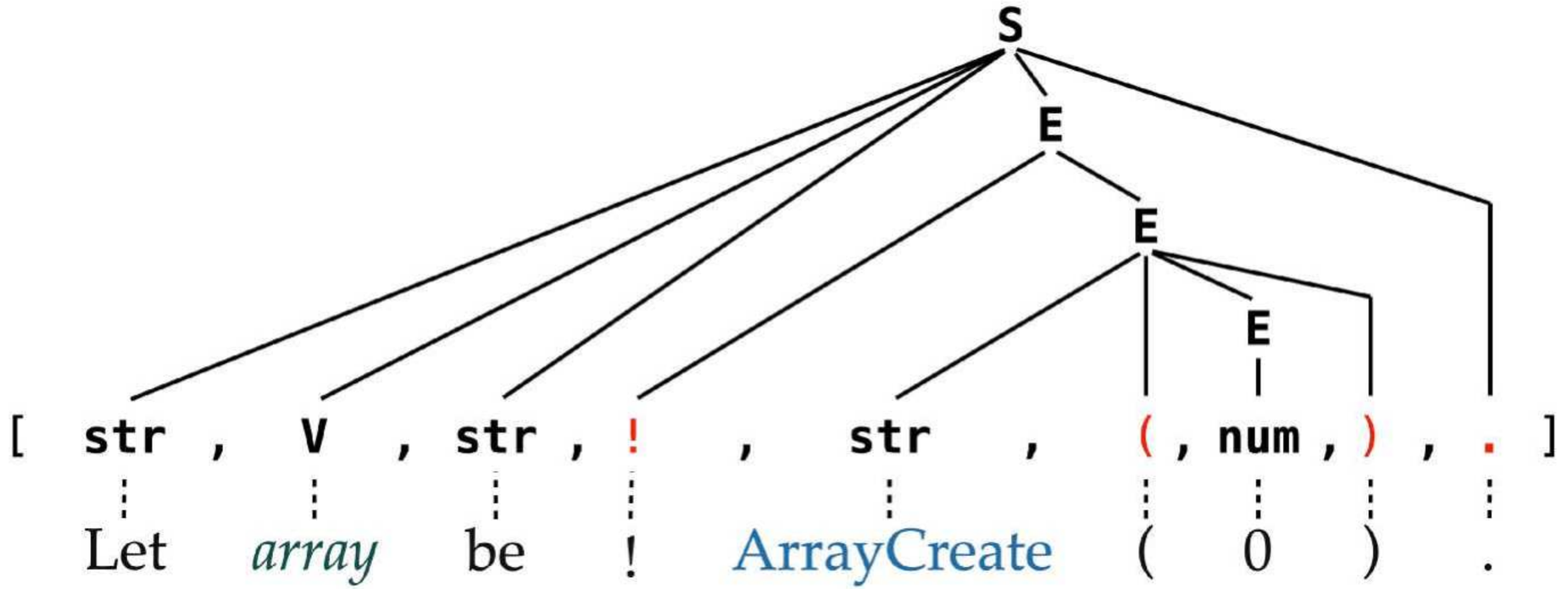


Parsing rules | Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ . ^^ ILet

E = // expressions
  ! E                  ^^ EAbruptCheck |
  str ~ ( ~ E ~ )      ^^ ECall        |
  num                  ^^ _.toDouble
```

Simplified compile rules



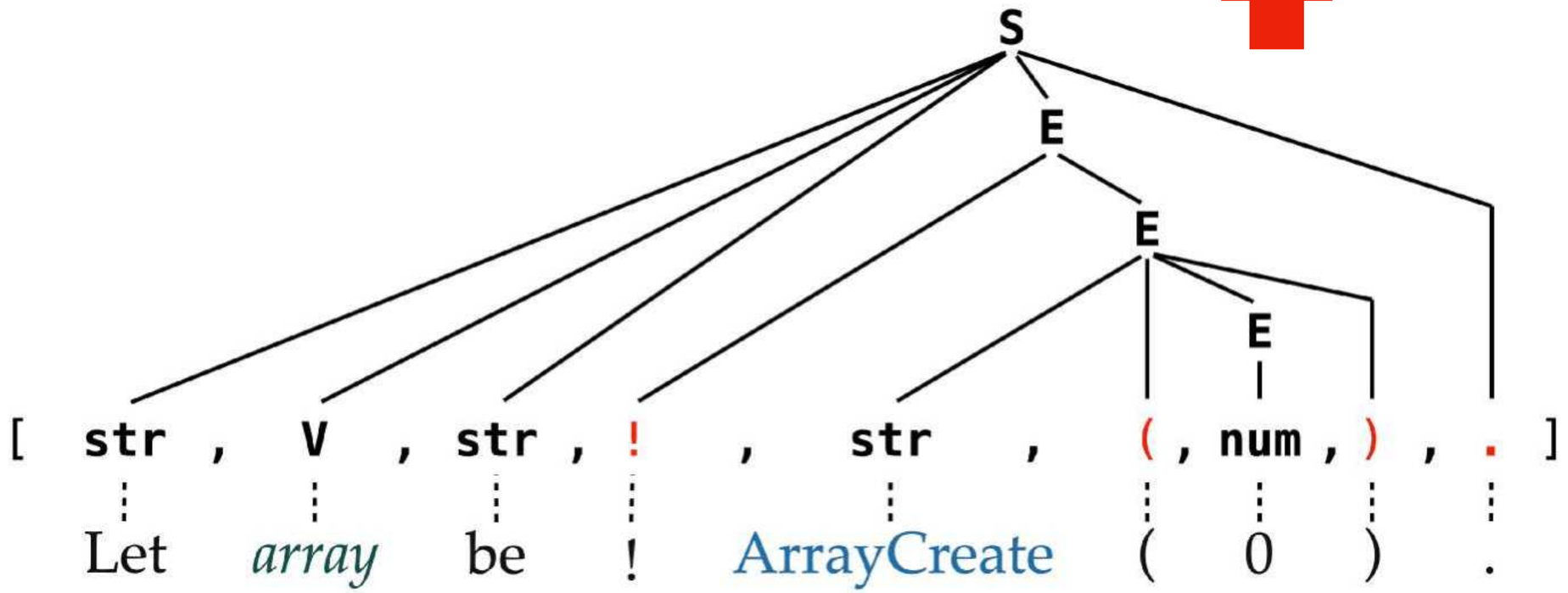
Parsing rules | Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ . ^^ ILet

E = // expressions
  ! E                  ^^ EAbruptCheck |
  str ~ ( ~ E ~ )      ^^ ECall        |
  num                  ^^ _.toDouble
```

Simplified compile rules

```
ILet(array, EAbruptCheck(
  ECall("ArrayCreate", 0)))
```



Parsing rules | Conversion Rules

```
S = // statements
  Let ~ V ~ be ~ E ~ . ^^ ILet

E = // expressions
  ! E                  ^^ EAbruptCheck |
  str ~ ( ~ E ~ )      ^^ ECall        |
  num                  ^^ _.toDouble
```

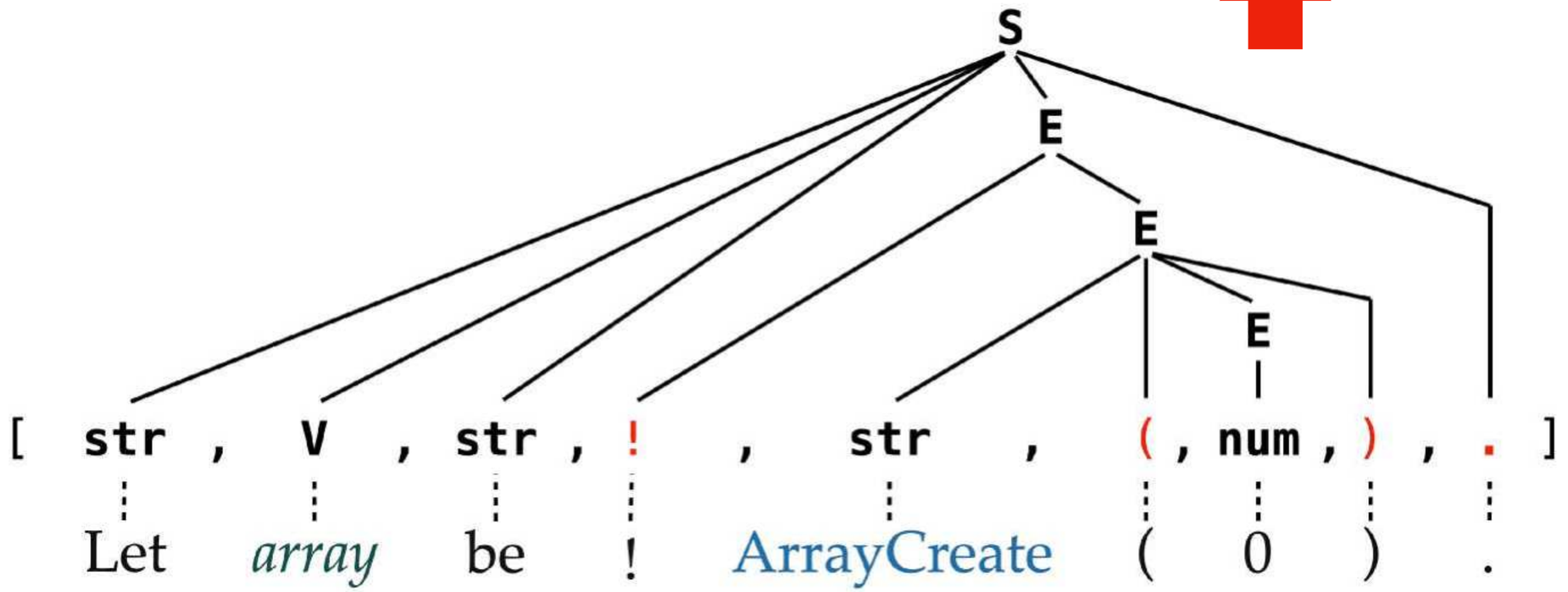
Simplified compile rules

```
let array = ! (ArrayCreate 0)
```



```
ILet(array, EAbruptCheck(
  ECall("ArrayCreate", 0)))
```

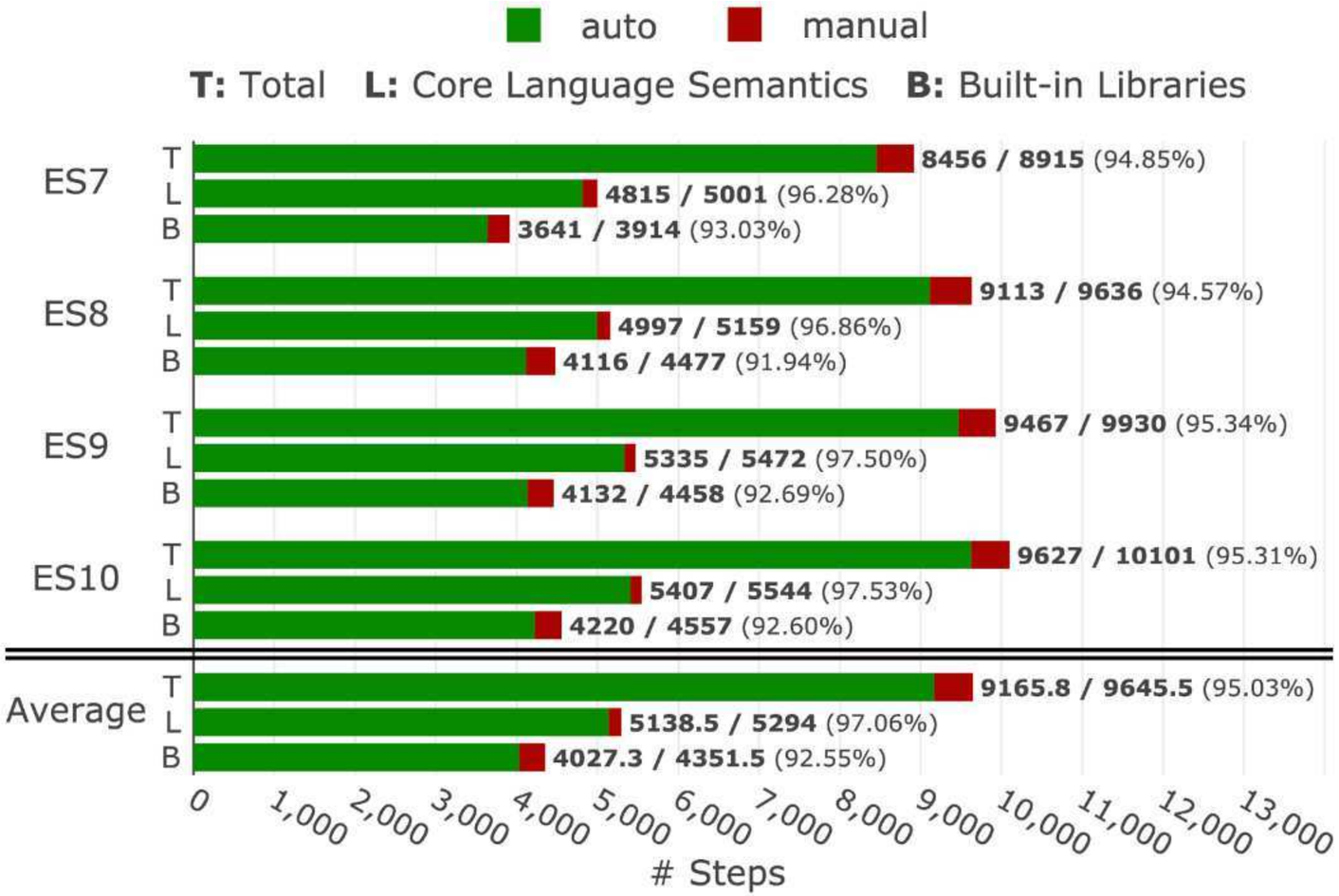




Semantics - Evaluation

The number of compile rules

Name	# Rules
Statment	21
Expression	27
Condition	16
Value	11
Type	34
Reference	9
Total	118

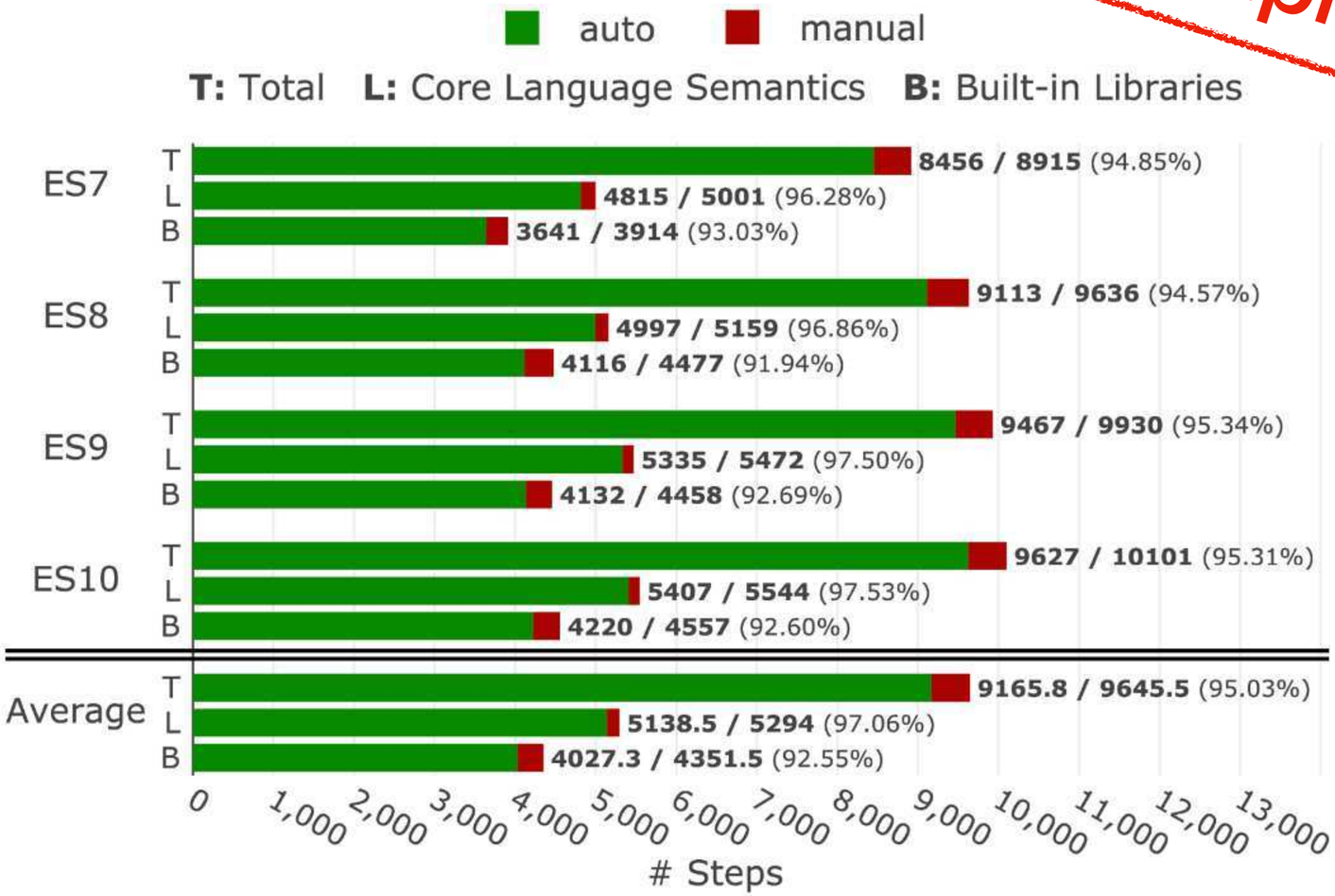


Semantics - Evaluation

≈ 95% Compiled

The number of compile rules

Name	# Rules
Statment	21
Expression	27
Condition	16
Value	11
Type	34
Reference	9
Total	118

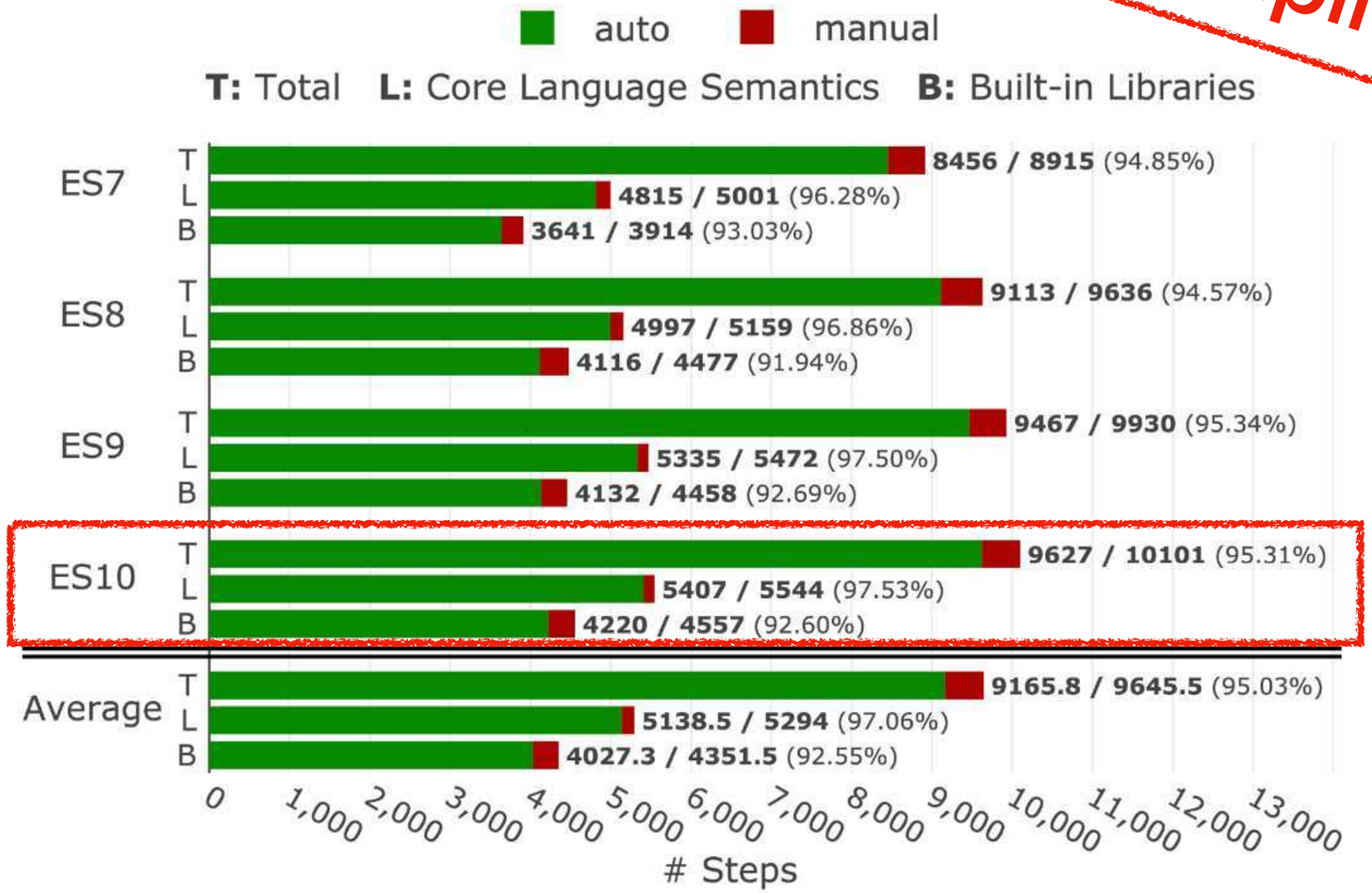


Semantics - Evaluation

≈ 95% Compiled

The number of compile rules

Name	# Rules
Statment	21
Expression	27
Condition	16
Value	11
Type	34
Reference	9
Total	118



Semantics - Evaluation

- **Test262** - Official ECMAScript test suite

16,355 / 18,064 (90.54%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
ES10-1	Iteration	Missing the `async-iterate` case in the assertion of **ForIn/OfHeadEvaluation**	X	2018-02-16	2020-03-25	768 days	1,116
ES10-2	Condition	Ambiguous grammar production for the dangling `else` problem in *IfStatement*	X	2015-06-01	TBD	TBD	1
ES10-3	String	Wrong use of the = operator in **StringGetOwnProperty**	X	2015-06-01	2020-05-07	1,802 days	7
ES10-4	Completion	Unhandling abrupt completion in **Abstract Equality Comparison**	X	2015-06-01	2020-04-28	1,793 days	9
ES10-5	Completion	Unhandling abrupt completion in **Evaluation** of *EqualityExpression*	O	2015-06-01	2019-05-02	1,431 days	2
ES10-6	Await	Passing a value of wrong type to the second parameter of **PromiseResolve**	O	2019-02-27	2019-04-13	45 days	1,294
ES10-7	Function	No semantics of **IsFunctionDefinition** for `function(...){...}`	O	2015-10-30	2020-01-18	1,541 days	306
ES10-8	Function	No semantics of **ExpectedArgumentCount** for the base case of *FormalParameters*	O	2016-11-02	2020-02-20	1,205 days	81
ES10-9	Iteration	Two semantics of **VarScopedDeclarations** for `for await(var x of e){...}`	O	2018-02-16	2019-10-11	602 days	0

292 / 303 (96.37%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
BigInt-1	Expression	Using the wrong variable `oldvalue` instead of `oldValue` in **Evaluation** of *UpdateExpression*	X	2019-09-27	2020-04-23	209 days	533
BigInt-2	Number	Using **ToInt32** instead of **ToUint32** in **Number::unsignedRightShift**	X	2019-09-27	2020-04-23	209 days	2
BigInt-3	Number	Unhandling BigInt values in the **Number** constructor	O	2019-09-27	2019-11-19	53 days	1

Semantics - Evaluation

- **Test262** - Official ECMAScript test suite

16,355 / 18,064 (90.54%)

9 bugs in ES10

18,064 / 18,064 (100.00%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
ES10-1	Iteration	Missing the `async-iterate` case in the assertion of **ForIn/OfHeadEvaluation**	X	2018-02-16	2020-03-25	768 days	1,116
ES10-2	Condition	Ambiguous grammar production for the dangling `else` problem in *IfStatement*	X	2015-06-01	TBD	TBD	1
ES10-3	String	Wrong use of the = operator in **StringGetOwnProperty**	X	2015-06-01	2020-05-07	1,802 days	7
ES10-4	Completion	Unhandling abrupt completion in **Abstract Equality Comparison**	X	2015-06-01	2020-04-28	1,793 days	9
ES10-5	Completion	Unhandling abrupt completion in **Evaluation** of *EqualityExpression*	O	2015-06-01	2019-05-02	1,431 days	2
ES10-6	Await	Passing a value of wrong type to the second parameter of **PromiseResolve**	O	2019-02-27	2019-04-13	45 days	1,294
ES10-7	Function	No semantics of **IsFunctionDefinition** for `function(...){...}`	O	2015-10-30	2020-01-18	1,541 days	306
ES10-8	Function	No semantics of **ExpectedArgumentCount** for the base case of *FormalParameters*	O	2016-11-02	2020-02-20	1,205 days	81
ES10-9	Iteration	Two semantics of **VarScopedDeclarations** for `for await(var x of e){...}`	O	2018-02-16	2019-10-11	602 days	0

292 / 303 (96.37%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
BigInt-1	Expression	Using the wrong variable `oldvalue` instead of `oldValue` in **Evaluation** of *UpdateExpression*	X	2019-09-27	2020-04-23	209 days	533
BigInt-2	Number	Using **ToInt32** instead of **ToUint32** in **Number::unsignedRightShift**	X	2019-09-27	2020-04-23	209 days	2
BigInt-3	Number	Unhandling BigInt values in the **Number** constructor	O	2019-09-27	2019-11-19	53 days	1

Semantics - Evaluation

- **Test262** - Official ECMAScript test suite

16,355 / 18,064 (90.54%)

9 bugs in ES10

18,064 / 18,064 (100.00%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
ES10-1	Iteration	Missing the `async-iterate` case in the assertion of **ForIn/OfHeadEvaluation**	X	2018-02-16	2020-03-25	768 days	1,116
ES10-2	Condition	Ambiguous grammar production for the dangling `else` problem in *IfStatement*	X	2015-06-01	TBD	TBD	1
ES10-3	String	Wrong use of the = operator in **StringGetOwnProperty**	X	2015-06-01	2020-05-07	1,802 days	7
ES10-4	Completion	Unhandling abrupt completion in **Abstract Equality Comparison**	X	2015-06-01	2020-04-28	1,793 days	9
ES10-5	Completion	Unhandling abrupt completion in **Evaluation** of *EqualityExpression*	O	2015-06-01	2019-05-02	1,431 days	2
ES10-6	Await	Passing a value of wrong type to the second parameter of **PromiseResolve**	O	2019-02-27	2019-04-13	45 days	1,294
ES10-7	Function	No semantics of **IsFunctionDefinition** for `function(...){...}`	O	2015-10-30	2020-01-18	1,541 days	306
ES10-8	Function	No semantics of **ExpectedArgumentCount** for the base case of *FormalParameters*	O	2016-11-02	2020-02-20	1,205 days	81
ES10-9	Iteration	Two semantics of **VarScopedDeclarations** for `for await(var x of e){...}`	O	2018-02-16	2019-10-11	602 days	0

292 / 303 (96.37%)

3 bugs in ES.Next

303 / 303 (100.00%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
BigInt-1	Expression	Using the wrong variable `oldvalue` instead of `oldValue` in **Evaluation** of *UpdateExpression*	X	2019-09-27	2020-04-23	209 days	533
BigInt-2	Number	Using **ToInt32** instead of **ToUint32** in **Number::unsignedRightShift**	X	2019-09-27	2020-04-23	209 days	2
BigInt-3	Number	Unhandling BigInt values in the **Number** constructor	O	2019-09-27	2019-11-19	53 days	1

Semantics - Evaluation

- **Test262** - Official ECMAScript test suite

All Tests Passed

16,355 / 18,064 (90.54%)

9 bugs in ES10

18,064 / 18,064 (100.00%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
ES10-1	Iteration	Missing the `async-iterate` case in the assertion of **ForIn/OfHeadEvaluation**	X	2018-02-16	2020-03-25	768 days	1,116
ES10-2	Condition	Ambiguous grammar production for the dangling `else` problem in *IfStatement*	X	2015-06-01	TBD	TBD	1
ES10-3	String	Wrong use of the = operator in **StringGetOwnProperty**	X	2015-06-01	2020-05-07	1,802 days	7
ES10-4	Completion	Unhandling abrupt completion in **Abstract Equality Comparison**	X	2015-06-01	2020-04-28	1,793 days	9
ES10-5	Completion	Unhandling abrupt completion in **Evaluation** of *EqualityExpression*	O	2015-06-01	2019-05-02	1,431 days	2
ES10-6	Await	Passing a value of wrong type to the second parameter of **PromiseResolve**	O	2019-02-27	2019-04-13	45 days	1,294
ES10-7	Function	No semantics of **IsFunctionDefinition** for `function(...){...}`	O	2015-10-30	2020-01-18	1,541 days	306
ES10-8	Function	No semantics of **ExpectedArgumentCount** for the base case of *FormalParameters*	O	2016-11-02	2020-02-20	1,205 days	81
ES10-9	Iteration	Two semantics of **VarScopedDeclarations** for `for await(var x of e){...}`	O	2018-02-16	2019-10-11	602 days	0

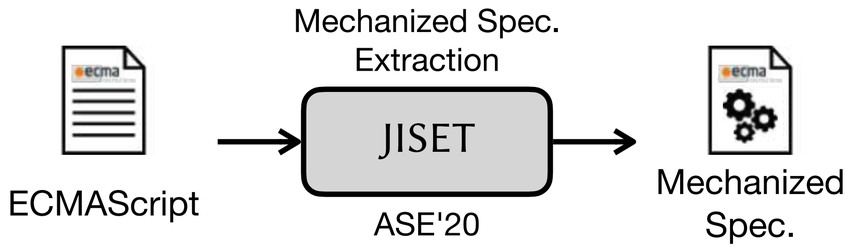
292 / 303 (96.37%)

3 bugs in ES.Next

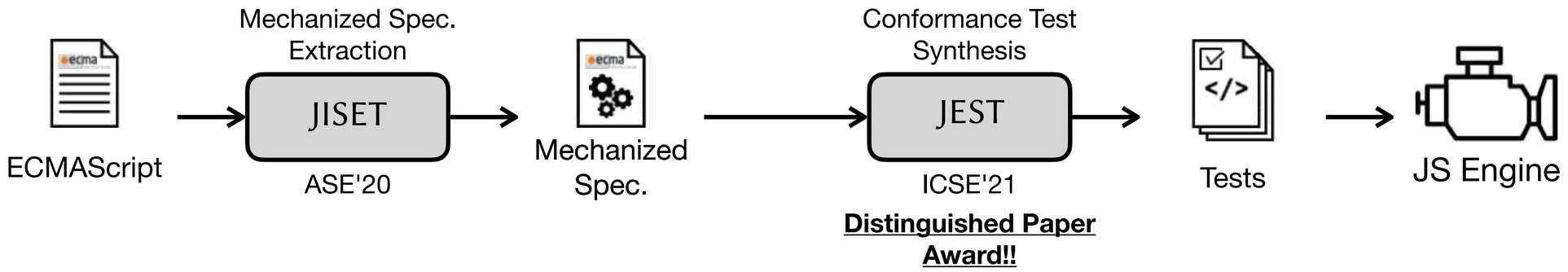
303 / 303 (100.00%)

Name	Feature	Description	Known	Created	Resolved	Existed	# Fails
BigInt-1	Expression	Using the wrong variable `oldvalue` instead of `oldValue` in **Evaluation** of *UpdateExpression*	X	2019-09-27	2020-04-23	209 days	533
BigInt-2	Number	Using **ToInt32** instead of **ToUint32** in **Number::unsignedRightShift**	X	2019-09-27	2020-04-23	209 days	2
BigInt-3	Number	Unhandling BigInt values in the **Number** constructor	O	2019-09-27	2019-11-19	53 days	1

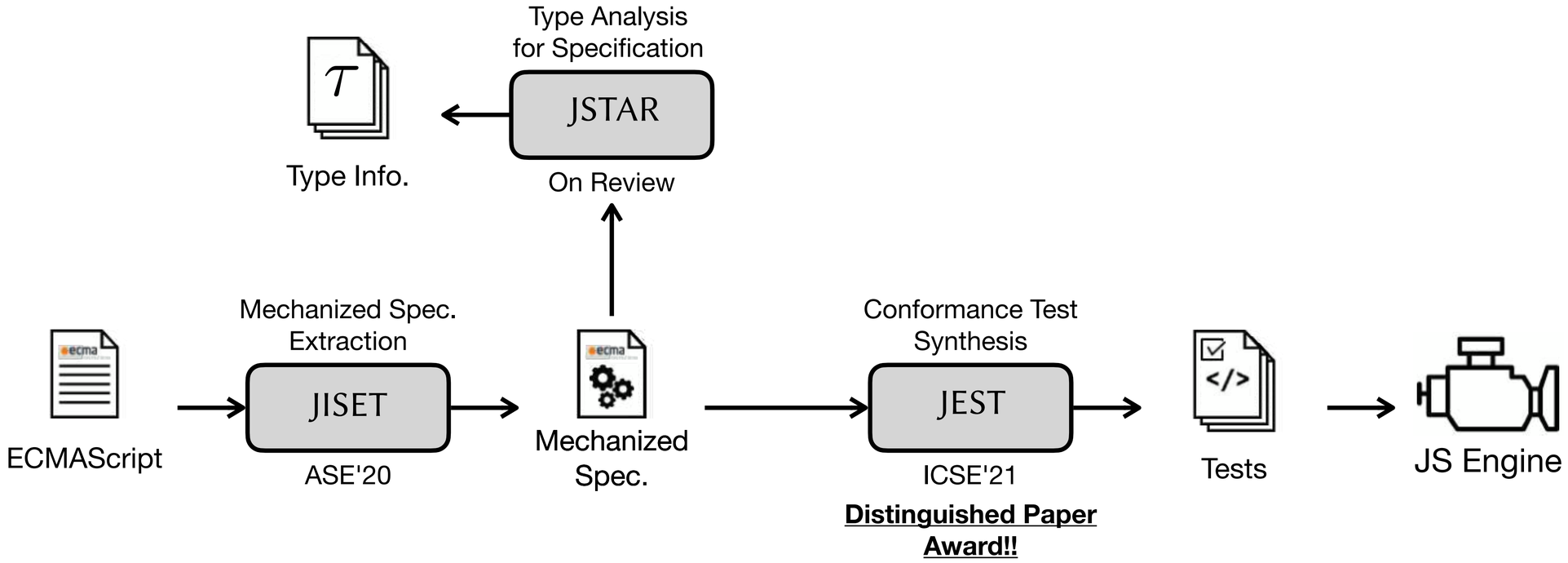
Future Work


Mechanized Spec.
Extraction
JISET
ECMAScript
ASE'20
Mechanized
Spec.

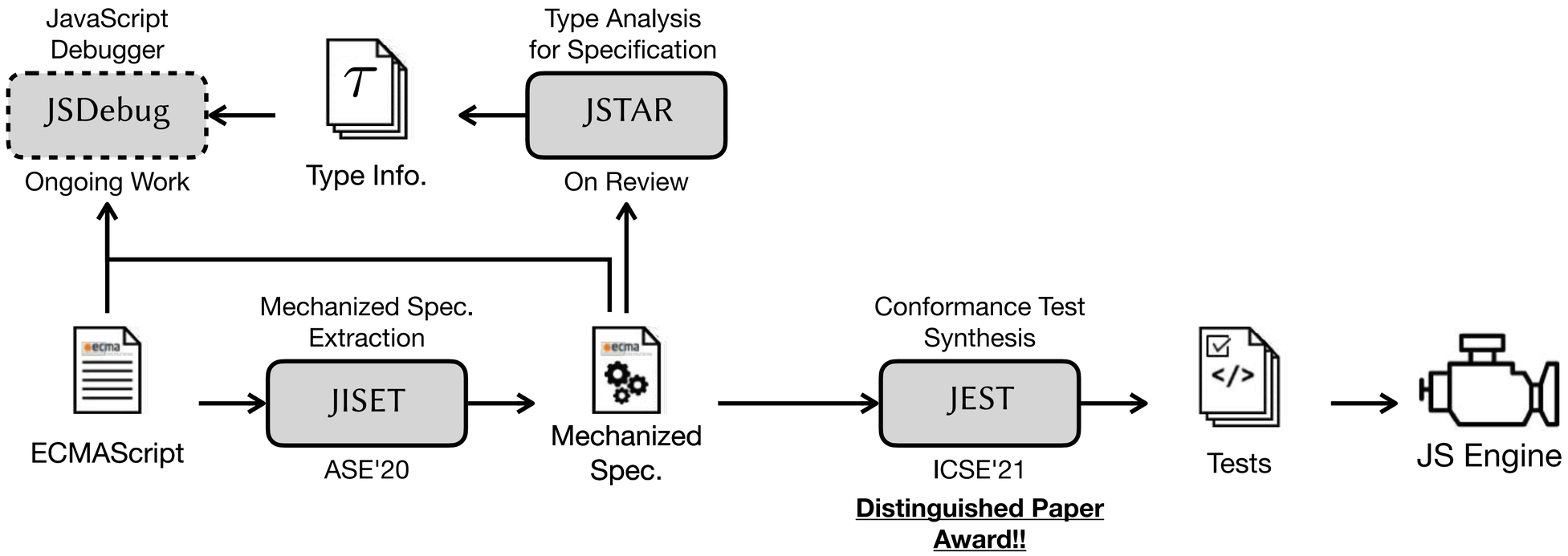
Future Work


Mechanized Spec. Extraction
JISET
ASE'20
ECMAScript
Mechanized Spec.
Conformance Test Synthesis
JEST
ICSE'21
Distinguished Paper Award!!
Tests
JS Engine

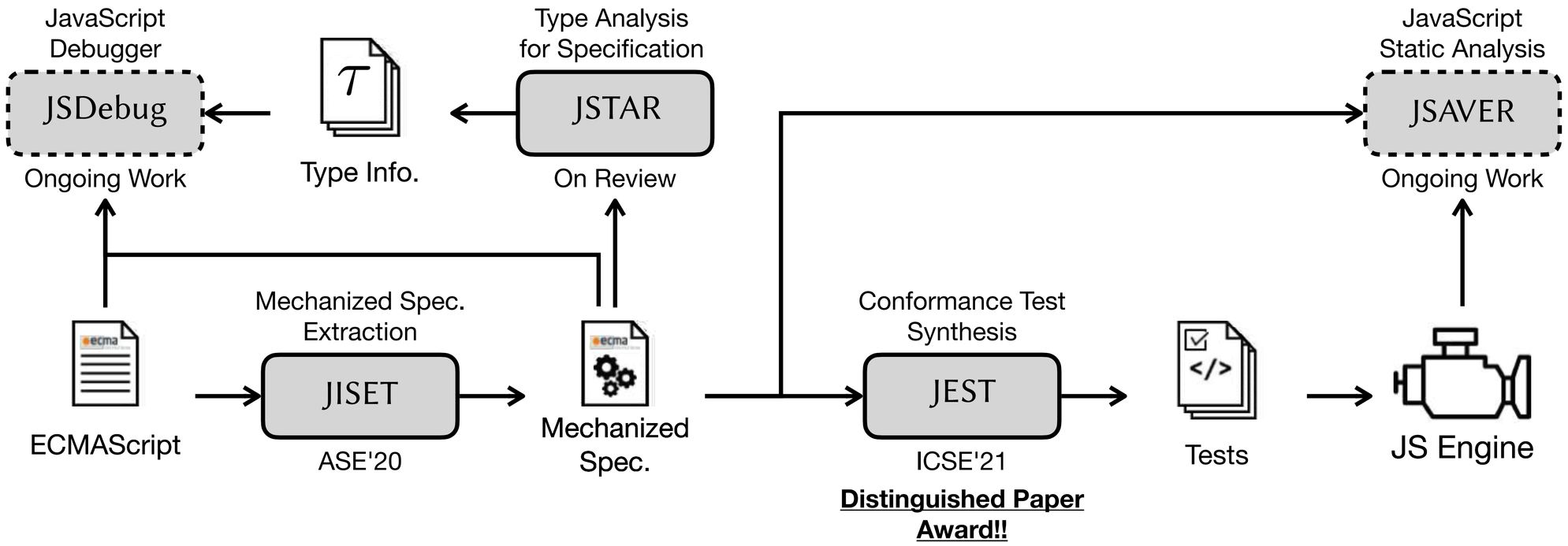
Future Work


Type Analysis
for Specification
JSTAR
On Review
Type Info.
Mechanized Spec.
Extraction
JISET
ASE'20
ECMAScript
Mechanized
Spec.
Conformance Test
Synthesis
JEST
ICSE'21
Distinguished Paper
Award!!
Tests
JS Engine

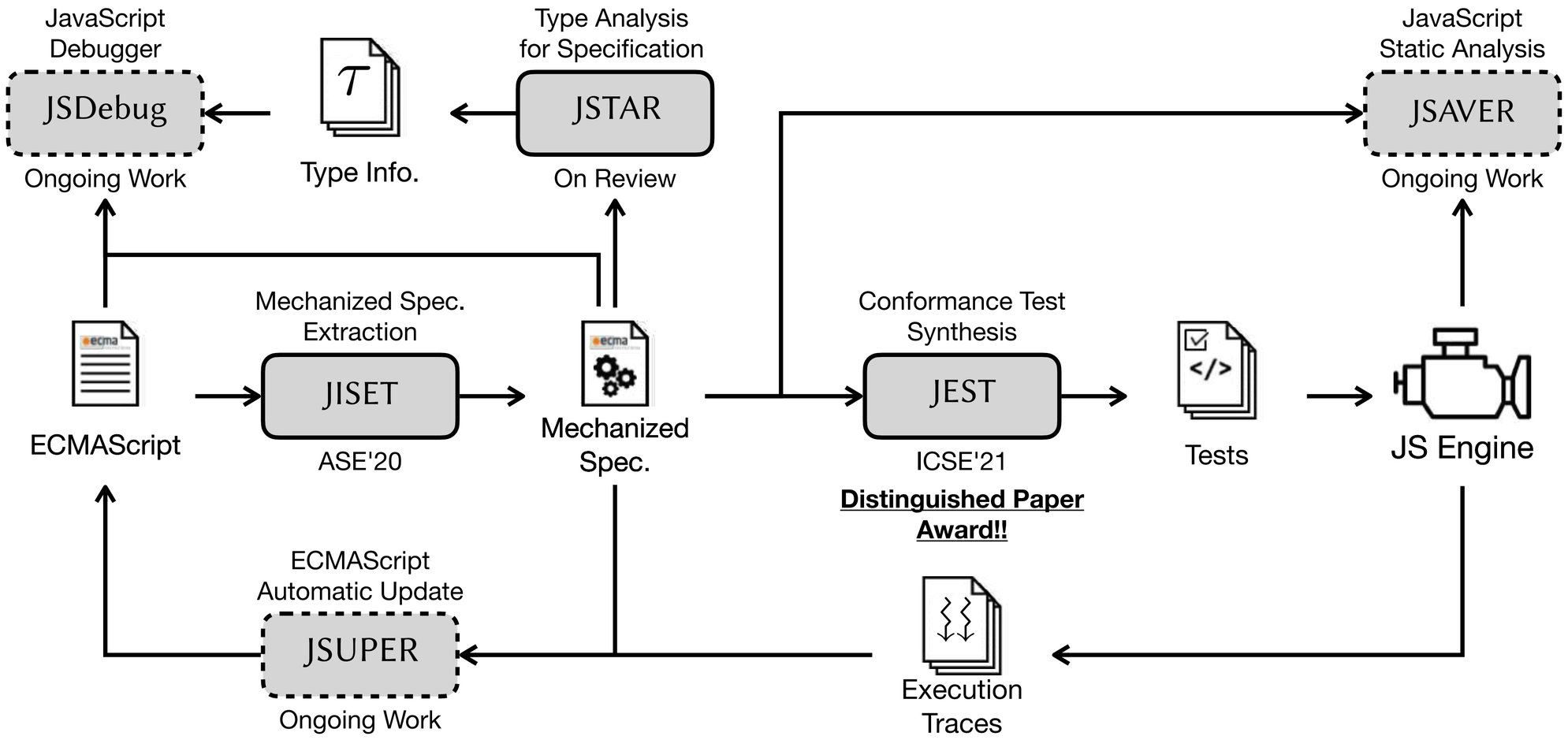
Future Work


JavaScript
Debugger
JSDebug
Ongoing Work
Type Info.
Type Analysis
for Specification
JSTAR
On Review
ECMAScript
Mechanized Spec.
Extraction
JISET
ASE'20
Mechanized
Spec.
Conformance Test
Synthesis
JEST
ICSE'21
Distinguished Paper
Award!!
Tests
JS Engine

Future Work


JavaScript
Debugger
JSDebug
Ongoing Work
Type Info.
Type Analysis
for Specification
JSTAR
On Review
JavaScript
Static Analysis
JSAVER
Ongoing Work
ECMAScript
Mechanized Spec.
Extraction
JISET
ASE'20
Mechanized
Spec.
Conformance Test
Synthesis
JEST
ICSE'21
Distinguished Paper
Award!!
Tests
JS Engine

Future Work


JavaScript
Debugger
JSDebug
Ongoing Work
Type Info.
Type Analysis
for Specification
JSTAR
On Review
JavaScript
Static Analysis
JSAVER
Ongoing Work
ECMAScript
Mechanized Spec.
Extraction
JISET
ASE'20
Mechanized
Spec.
Conformance Test
Synthesis
JEST
ICSE'21
Distinguished Paper
Award!!
Tests
JS Engine
ECMAScript
Automatic Update
JSUPER
Ongoing Work
Execution
Traces